edx Phot1x report

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***Abstract*—This report outlines the course project of the UBCx Phot1x - Silicon Photonics Design, Fabrication and Data Analysis course. Several unblanaced Mach-Zehnder interferometers are designed and fabricated on a Silicon-on-Insulator platform. Different path length differences of the interferometers allow us to measure the free spectral range and infer the group index of the photonic circuits.**

I. Introduction

A fundamental building block in optics is the Mach-Zehnder-interferometer (MZI), which exploits optical interference between two paths to achieve filtering, modulation and phase measurements. Integrated photonics reduces the size of such components significantly and enables scalable devices for applications in optical communications, sensing, signal processing and optical computation.

An important parameter for wave propagation in integrated waveguides is the group index, which determines teh velocity of optical pulses and impacts the performance of the on-chip components. Accurate knowledge of the group index is therefore essential for both the design and the characterization of photonic circuits. A practical method to extract the group index is through the spectral response of an unbalanced MZI. The periodic transmission spectrum characterized by its free spectral range (FSR) and is directly related to the group index and the path length difference between the interferometer arms.

In this project of the edx Phot1x course, a set of MZIs is designed on a silicon photonics platform. The waveguide height is fixed at 220 nm and the nominal width is 500 nm, so that mainly the TE fundamental modes is guided. To investigate the relationship between the armlength difference of the interferometer and the spectral response, we design six MZIs with path length differences (ΔL) of 30 μm, 60 μm, 100 μm, 150 μm, 250 μm, 500 μm. This allows us to study the varying free spectral ranges within the constrains of the measurement bandwidth of 50 nm. This ensures that the group index can be reliably extracted.

The objective of this work is to model, simulate, and analyze these interferometer designs, and to establish a methodology for extracting the waveguide group index from both simulated and experimental transmission spectra.

II. Theory

A. Mach-Zehnder Interferometer

The interferometer consists of two optical paths that are split and recombined. The output intensity is determined by the phase difference of the two arms. The propagation constant of light traveling thorough the waveguide is

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda} \ ,$$

where n_{eff} is the effective index of refraction and λ the wavelength of the light.

If the two arms of the MZI have lengths L_1 and L_2 and the path length difference is $\Delta L = L - 2 - L_1$, we can derive the phase difference:

$$\Delta\phi = \beta\Delta L = \frac{2\pi n_{\text{eff}}\Delta L}{\lambda}$$

If a single electric field E_{in} is split and recombined, the two split fields interfere on the beam splitter/combiner. The field at the output is then:

$$E_{out} = \frac{E_{in}\mathrm{e}^{\phi} + \mathrm{E_{in}}}{2}$$

and the resulting output intensity - also called the transfer function - is

$$I_{out} = T(\lambda) = \frac{1}{2} I_{in} \left(1 + \cos\left(\Delta\phi\right)\right) \ .$$

If the combining beam splitter has two output ports, the second port has a $pi/2$ phase shift.

B. Free Spectral Range (FSR)

We see constructive interference (in the case of two outputs, the other port shows destructive interference) when the phase difference is a multiple of 2π:

$$\Delta\phi = 2\pi m, \quad m \in \mathbb{Z} \ .$$

For two adjacent maxima at wavelengths (λ) and ($\lambda + \Delta\lambda$), the phase condition leads to the definition of the free spectral range (FSR).

$$\Delta\lambda = \frac{\lambda^2}{n_g \Delta L}$$

where (n_g) is the group index of the waveguide.

C. Group index

The group index that was introduced in the last equation is defined as:

$$n_g = n_{\text{eff}} - \lambda \frac{\partial n_{\text{eff}}}{\partial \lambda}$$

The group iindex accounts for dispersion and describes how the phase velocity changes with wavelength. It determines the spacing of the interference fringes in the MZI spectrum. We can rearrange the expression for the FSR:

$$n_g = \frac{\lambda^2}{\Delta\lambda\delta L} \ .$$

Therefore, by measuring the FSR ($\Delta\lambda$) from the transmission spectrum and the path length difference (ΔL) defined by the layout design, the group index can be determined.

III. Modelling and Simulation

A. Waveguide Geometry and Mode Analysis

The MZI is based on a standard silicon-on-insulator (SOI) strip waveguide, which is designed for the wavelengths in the telecom C-band (1550 nm). The geometry is optimized to support quasi-single-mode propagation in the quasi-TE polarization. The design parameters are summarized as follows:

- Core material: Silicon (Si), refractive index n $\approx$ 3.47 at 1550 nm
- Cladding: Silicon dioxide (SiO2) for both, upper and lower cladding (n $\approx$ 1.44)
- Waveguide width: 500 nm
- Waveguide height: 220 nm
- Polarization: Quasi-TE mode

The chosen geometry has strong confinement in the silicon core and evanescent decay at the boundaries. It only supports the fundamental mode. A simulation of the fundamental TE mode was done with Lumerical MODE Solutions and can be seen in Fig. 1.

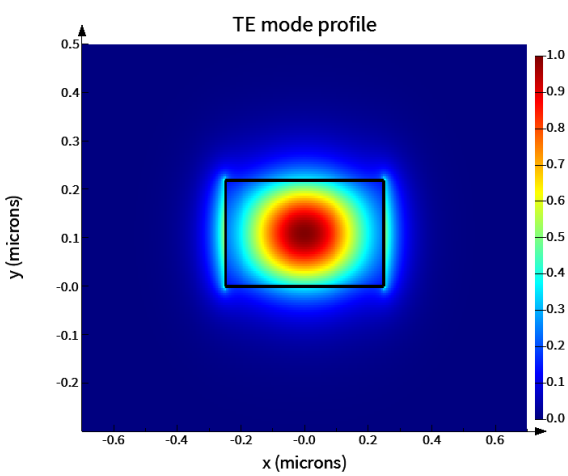

Figure 1. TE mode profile simulation for a waveguide of h=220μm, w=500μm.

B. Effective Index and Group Index

To get a better understanding of the dispersive properties of our designed waveguide, we computed the effective index (n_{eff}) and the group index (n_g) as a function of wavelength (See Fig.2,3). The simulations indicate a group index of $\approx$ 4.197, which serves as a reference to validate the interferometric extraction method.

C. Compact Model of the Waveguide

For circuit-level simulation, we can model the wavelength-dependent effective index by fitting a second-order polynomial:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2 \ .$$

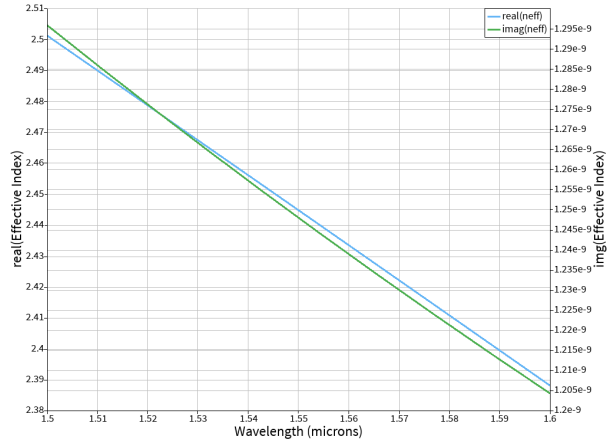

Figure 2. Simulation of the effective index for a waveguide of h=220μm, w=500μm over the telecom C-band range.

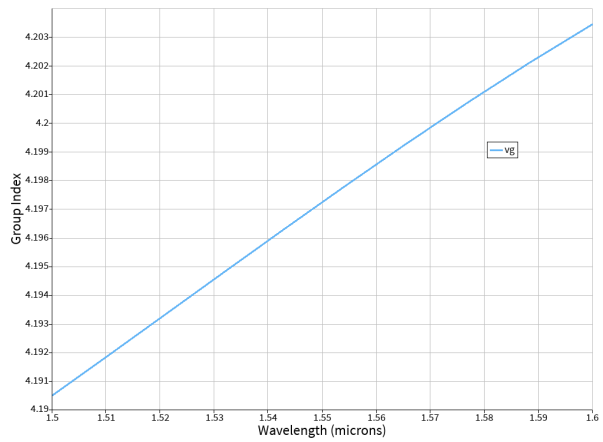

Figure 3. Simulation of the group index for a waveguide of h=220μm, w=500μm over the telecom C-band range.

By fitting our simulated wavelength sweep in Lumerical MODE, we can directly extract the coefficients. We get as a result for the effective index:

$$n_{\text{eff}}(\lambda) = 2.447 - 1.133(\lambda - 1.55) + 0.044(\lambda - 1.55)^2 \ .$$

D. Design verification and extraction of group index

In the theory section, we presented the wavelength dependent expression for the interferometer transfer function and also the FSR. The relationship between the path length difference of the interferometer arms and the group index can be used to experimentally extract the value of the group index from different MZIs. As a benchmark and to compare our experimental results later on, we can calculate the expected FSR with the simulated value for n_g. We can also use Lumerical INTERCONNECT to simulate our designed MZI. A transmission spectrum for the MZI with arm length difference of 100 μm is shown in Fig. 4. The two plotted curves are the two outputs of the MZI. The plotted curves are the result of the characteristic interference pattern and the grating coupler efficiency over the wavelength range.

The chosen values for the path difference are in Table 1, together with the expected and simulated group index values.

Δ L [μ m]	Exp. FSR [nm]	Sim. FSR [nm]
30	19.08	18.71
60	9.54	9.50
100	5.72	5.75
150	3.82	3.85
250	2.29	2.30
500	1.14	1.14

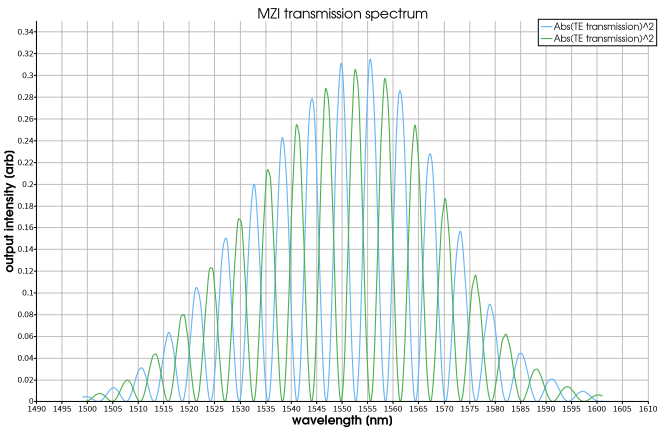

Figure 4. Measured transmission spectrum on a Mach-Zehnder Interferometer with a path length difference of 100 microns.

IV. Conclusion

The conclusion goes here.

V. Acknowledgements

(edit according to your use).

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